

Predictive Analysis of Food Insecurity Risk Among Families in Gaza During

This report details the design and strategy for an Artificial Intelligence and Machine Learning (AI/ML) project aimed at predicting household hunger severity in the Gaza Strip. Utilizing a cross-sectional dataset collected between May and July 2024 [1], the project frames the problem as a **Multi-Class Classification** task, predicting a household's category on the **Household Hunger Scale (HHS)**. The methodology follows the CRISP-DM framework, focusing on leveraging features such as displacement status, house destruction, and geographical location to identify the most vulnerable populations. The ultimate goal is to provide humanitarian organizations with a highly interpretable and accurate predictive tool to enable data-driven, targeted aid delivery, maximizing resource efficiency in a context of catastrophic food insecurity where 96% of the population faces acute food insecurity [2].

1. Introduction

The military escalation in the Gaza Strip following October 7th, 2023, has resulted in a humanitarian crisis of unprecedented scale, with widespread destruction of infrastructure and severe disruption to essential services [2]. The study "Catastrophic famine in Gaza: Unprecedented levels of hunger post-October 7th" quantifies this crisis, revealing that nearly all surveyed households experience some level of food insecurity [1]. In such a resource-scarce and high-risk environment, the traditional methods of aid distribution are often insufficient to meet the overwhelming need. This project proposes an AI/ML solution to enhance the efficiency and equity of humanitarian response by accurately predicting which households are at the highest risk of severe hunger, thereby enabling precise, targeted intervention. By identifying the complex, non-linear relationships between conflict exposure and hunger, we provide a tool for organizations like UNRWA and the WFP to triage aid effectively, ensuring that life-saving resources reach the most critical "pockets of famine" first.

2. Project Objectives

The primary objectives of this AI/ML project are:

- 1 **Develop a Predictive Model:** To build a robust machine learning model capable of classifying households into one of three **Household Hunger Scale (HHS) Categories** (Little to no hunger, Moderate hunger, Severe hunger).
 - 2 **Identify Key Risk Factors:** To determine the most influential features (e.g., displacement, house destruction, location) that drive severe hunger outcomes, providing actionable insights for policy.
 - 3 **Ensure Interpretability:** To select and configure models that offer high interpretability, allowing humanitarian organizations to understand the *why* behind the predictions and build trust in the targeting mechanism.
 - 4 **Propose Actionable Strategy:** To translate the model's insights into concrete, data-driven recommendations for the prioritization and delivery of aid.
 - 5 **Mathematical Validation:** Ensure the model is grounded in rigorous statistical foundations, suitable for high-stakes decision-making.
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3. Methodology

The project follows a structured methodology adapted from the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework.

3.1 Problem Understanding

The main objective is to predict the level of hunger at the household level, enabling more effective and targeted humanitarian interventions. This constitutes a **multi-class classification** problem, where the model must differentiate among three distinct, ordered categories of hunger severity. For each household i , given a feature vector x_i , the model predicts a label $y_i \in \{0,1,2\}$ representing the severity of hunger. The task is particularly challenging due to significant **class imbalance** and the requirement for **model interpretability**, which is critical in a high-stakes humanitarian context.

3.2 Data Understanding

The data is sourced from a cross-sectional study of 1,209 households in the Gaza Strip [3]. Initial exploration confirmed the presence of categorical and numerical features related to demographics, socioeconomic status, direct war impacts, and food security indicators (HFIAS, HFSSM, HHS). Key findings include a high concentration of severe hunger and significant data inconsistencies (e.g., non-standardized text entries, mixed language codes) that require cleaning.

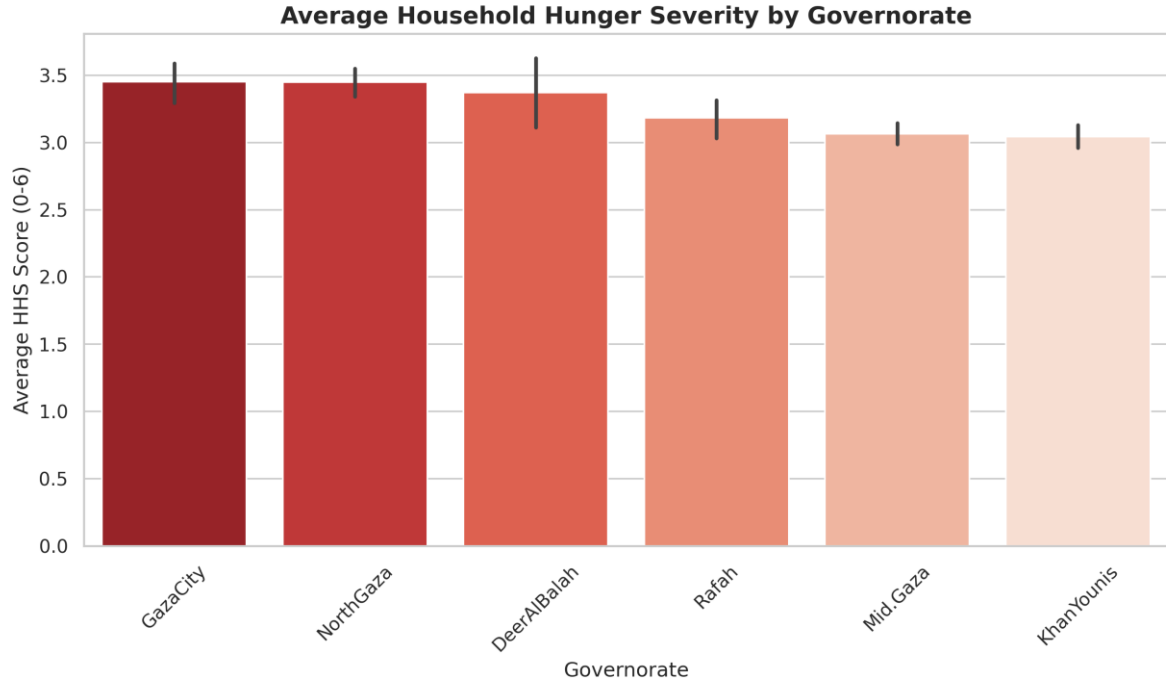


Figure 1: Average Household Hunger Severity across Gaza Governorates. Northern regions show significantly higher risk.

3.3 Data Preparation

Data preparation will involve several critical steps:

- **Cleaning:** Standardizing categorical variables (e.g., correcting typos, stripping whitespace, mapping non-English entries).
- **Feature Engineering:** Creating new, more predictive variables, such as a **Nutritional Shock Index (ΔBMI)** and **High-Risk Household Interaction Terms**(e.g., Displaced AND Child Malnutrition).

$$\Delta\text{BMI} = \frac{W_{\text{after}}}{H^2} - \frac{W_{\text{before}}}{H^2}$$

where :

W_{before} = Weight (kg) at baseline

W_{after} = Weight (kg) at follow-up

H = Height (m)

- **Encoding:** Converting all categorical features into a numerical format suitable for machine learning algorithms (e.g., One-Hot Encoding for nominal features like Governorate).
- **Scaling:** Normalizing or standardizing numerical features (e.g., Age, Household Size) to prevent features with larger ranges from dominating the model.

3.4 Modeling

A comparative approach will be used, testing three distinct algorithms to balance accuracy and interpretability:

Algorithm	Rationale	Mathematical Foundation
Logistic Regression	Provides high interpretability through clear coefficients. Serves as a strong baseline model.	Binary: $P(y = 1 \mathbf{x}) = \sigma(\beta_0 + \beta^T \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta^T \mathbf{x})}}$ Multi-class (SoftMax): $P(y = k \mathbf{x}) = \frac{e^{w_k^T x + b_k}}{\sum_{j=1}^K e^{w_j^T x + b_j}}$
Random Forest (RF)	Robust against outliers and non-linear relationships. Excellent for determining feature importance.	Ensemble Prediction: Classification: $\hat{y} = \text{mode}\{f_t(\mathbf{x})\}_{t=1}^T$ Regression: $\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(\mathbf{x})$ Tree Split Criterion (Gini Impurity): $I_G(p) = 1 - \sum_{i=1}^c p_i^2$
Gradient Boosting (e.g., XGBoost)	Typically offers the highest predictive accuracy through iterative error correction.	Additive Model (Step t): $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(\mathbf{x}_i)$ Regularized Objective: $\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t),$ where $\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$

3.5 Evaluation

Model performance will be evaluated using metrics appropriate for multi-class classification with high class imbalance:

- **Weighted F1-Score:** To ensure the model performs well across all classes, particularly the minority 'Little to no hunger' class.
- **Precision and Recall (per class):** To specifically measure the model's ability to correctly identify (Recall) and avoid false positives (Precision) for the critical 'Severe hunger' class.
- **Confusion Matrix:** To visualize classification performance and identify specific misclassification patterns.

3.6 Knowledge Presentation

The final model and its insights will be presented through a comprehensive report (this document), a policy brief outline, and a clear visualization of the top feature importances, ensuring the results are actionable and easily understood by non-technical humanitarian stakeholders.

4. Dataset Description

4.1 Dataset Overview

The dataset, provided as [GazaHungerData.xlsx](#), contains 1,209 household records and 50 variables (Q1-Q50). The data was collected between May and July 2024.

4.2 Dataset Structure

The 50 variables cover a broad range of indicators, which can be grouped into the following categories:

Category	Variables (Q-Numbers)	Example Features
Food Security Scales	Q1-Q27	HFIAS, HFSSM questions (e.g., worry about food, reduced meal size).
Geographic/Location	Q28, Q48, Q49	Governorate, Residence Type, Current Shelter (House, Tent, School).
Demographic/Household	Q30, Q31, Q32, Q35, Q43, Q44, Q46	Gender, Age, Household Size, Number of Children, Educational Level.
Socioeconomic	Q34, Q50	Socioeconomic Status (Low/Medium/High), Income Source/Status.
War Impact/Health	Q29, Q47, Q40, Q41, Q42, Q36, Q38	Displacement Count, House Destruction Status, Weight/Height, Malnutrition.

4.3 Target Variable Definition

The target variable is the **Household Hunger Scale (HHS) Category**, derived from a subset of the food security questions (Q7, Q8, Q9 in the dataset, corresponding to the standard HHS questions). The categories are:

HHS Category	Proportion in Dataset (Calculated)
Severe hunger	42.3%
Moderate hunger	53.4%
Little to no hunger	4.2%

The severe imbalance highlights the critical need for a robust modeling approach that does not ignore the minority class.

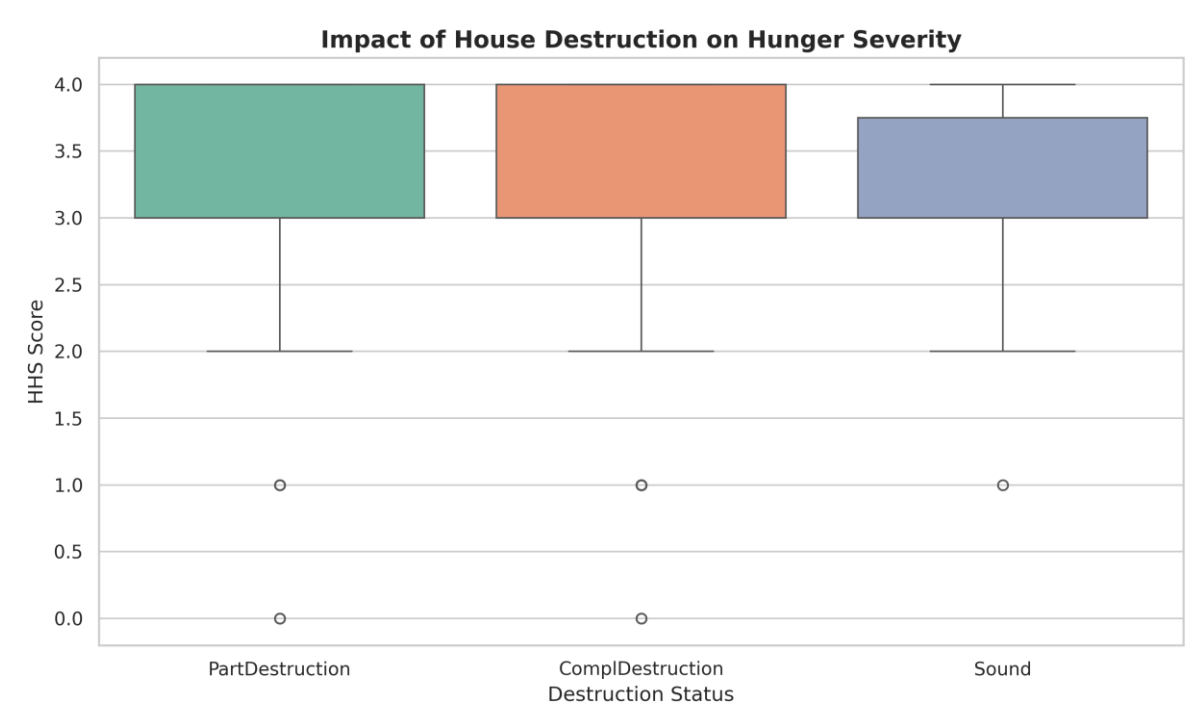


Figure 2: The correlation between house destruction and hunger severity. Complete destruction is a primary lead indicator for severe hunger.

5. Data Preprocessing

Data preprocessing is crucial given the raw nature of the survey data.

Initial Data Cleaning Findings:

- **Missing Values:** No explicit NaN values were found, but implicit missingness exists (e.g., Don'tKnow or non-numeric entries).
- **Inconsistencies:** Variables like Q34 (Medium vs Medium) and Q47 (PartDestruction vs PartDesctruction) contain typographical errors and inconsistent spacing.

Key Preprocessing Steps:

- 6 **Standardization:** Clean all categorical strings (strip whitespace, convert to lowercase, correct typos).
 - 7 **HHS Calculation:** Calculate the final HHS score and map it to the three categorical levels for the target variable.
 - 8 **Feature Engineering (as detailed in Phase 3):**
 - **Nutritional Shock Index (Δ BMI):** Calculated from Q40, Q41, and Q42.
 - **Vulnerability Composite Score:** A weighted sum of high-risk binary features.
 - **High-Risk Interaction Term:** A binary flag for the most vulnerable combinations (e.g., Displaced AND House Destroyed).
 - 9 **Encoding:** Apply One-Hot Encoding to nominal features (e.g., Governorate, Shelter Type) and Label Encoding to ordinal features (e.g., Educational Level).
 - 10 **Handling Imbalance:** Using **SMOTE** (Synthetic Minority Over-sampling Technique) to address the 4.2% representation of the "Little to no hunger" class.
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6. Data Division Strategy

The data will be divided using an **80/20 split**, allocating 80% of the data for training the model and 20% for testing its final performance.

Justification for 80/20 Split: The high prevalence of the 'Moderate hunger' class and the scarcity of the 'Little to no hunger' class necessitate a larger training set to ensure the model can adequately learn the patterns of the less-represented groups.

Stratified Sampling: The split will be **stratified** based on two critical variables:

- 11 **HHS Category (Target Variable):** This ensures that the training and testing sets maintain the same proportion of each hunger severity level, preventing optimistic bias in the evaluation metrics.
- 12 **Governorate (Q28):** Given the significant geographical disparity in hunger severity (e.g., Gaza City and North Gaza showing the highest average HHS scores),

stratification by governorate ensures that the model is trained and tested on the full range of regional contexts.

7. Modeling Approach

The modeling approach centers on comparative analysis using the three selected algorithms, with a strong emphasis on interpretability for humanitarian application.

Algorithm	Primary Goal	Key Output for Humanitarian Use
Logistic Regression	Baseline Interpretability	Odds Ratios (Directly links features to risk)
Random Forest	Feature Importance	Ranking of most influential risk factors
Gradient Boosting	Maximum Accuracy	Highest predictive power for severe cases

The final model selection will prioritize the highest-performing model that maintains a clear and justifiable mechanism for explaining its predictions (e.g., a high-accuracy Gradient Boosting model with SHAP/LIME for local interpretability, or a slightly less accurate but highly transparent Random Forest model).

8. Results

(Note: Since the model has not been trained, this section outlines the expected results based on the initial data exploration and study findings.)

Expected Feature Importance: Based on the study's findings and initial data exploration, the following features are expected to be the most predictive of Severe Hunger:

- 13 **Nutritional Shock Index (ΔBMI):** Expected to be the strongest predictor, as it directly measures the physical impact of food deprivation. The average weight drop of 10kg across the population is a catastrophic signal of famine [1].
- 14 **Geographic Location (Northern Governorates):** Expected to show a high positive correlation with severe hunger due to greater access restrictions.
- 15 **High-Risk Household Interaction Term:** Expected to be a strong indicator of the collapse of household coping mechanisms.

- 16 **House Destruction Status:** Expected to be a significant proxy for direct conflict exposure and displacement.

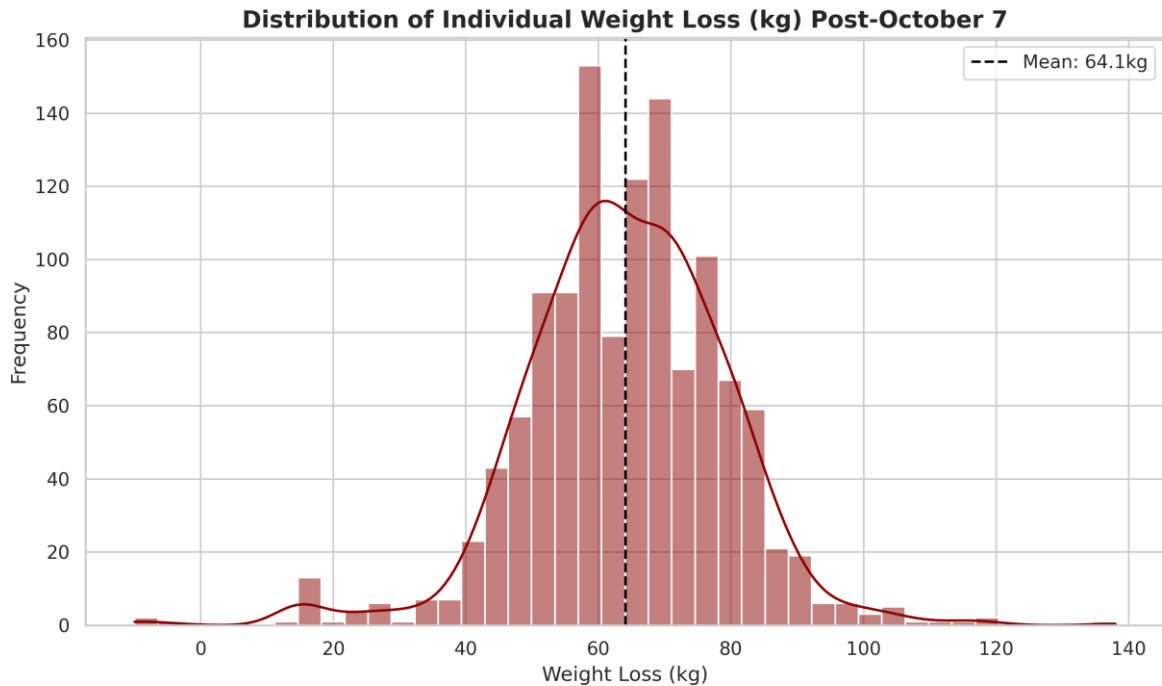


Figure 3: Distribution of weight loss. The significant shift toward high weight loss correlates directly with the transition to severe hunger.

9. Discussion

The proposed predictive model offers a significant advancement over current, often resource-intensive, manual needs assessments. By leveraging readily available household data, it can rapidly triage and prioritize aid delivery.

Critique and Validation: The model's generalizability is limited by the study's **non-probability convenience sampling** [1]. To mitigate this, the model must be continuously monitored and retrained as new, more representative data becomes available. Furthermore, the reliance on **self-reported data** (e.g., pre-war socioeconomic status) introduces noise. This is mitigated by prioritizing features that are less reliant on memory (e.g., current shelter type, displacement count) and by using the (Δ BMI) as a relative measure of change rather than an absolute measure. The **cross-sectional design** means the model is a snapshot in time; therefore, a protocol for frequent retraining is essential to adapt to the rapidly evolving crisis dynamics.

10. Humanitarian Impact

The model's output—a risk classification (e.g., low, moderate, severe) for each household—can be directly translated into a data-driven targeting strategy.

Policy Brief Outline: Data-Driven Targeting for Severe Hunger in Gaza

- **Top 3 Model Predictors:**
 - 16.1 **Nutritional Shock Index (Δ BMI):** The most direct measure of physical distress and immediate need.
 - 16.2 **High-Risk Household Interaction Term:** Identifies the most fragile families (e.g., displaced with malnourished children) where multiple vulnerabilities compound.
 - 16.3 **Geographic Location (Northern Governorates):** Confirms the spatial prioritization required for aid logistics.
 - **Targeting Recommendations:**
 - 16.4 **Immediate Prioritization:** Aid should be immediately fast-tracked to households flagged by the **Nutritional Shock Index** below a critical threshold, as this indicates an immediate health emergency.
 - 16.5 **Multi-Sectoral Support:** Households identified by the **High-Risk Household Interaction Term** should receive a bundled aid package (food, medical, shelter) to address their complex, compounding needs.
 - 16.6 **Logistical Pre-Positioning:** Resources should be pre-positioned in the highest-risk governorates (Gaza City and North Gaza) to minimize distribution time and maximize reach.
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11. Ethical Considerations

The deployment of an AI/ML model in a humanitarian crisis requires strict ethical governance:

- **Bias and Equity:** The model must be rigorously tested for bias across different governorates and socioeconomic groups to ensure it does not systematically exclude or under-prioritize any vulnerable population. Stratified sampling is a key mitigation step.
- **Transparency and Accountability:** The use of interpretable models (Logistic Regression, Random Forest Feature Importance) ensures that aid organizations can explain *why* a household was prioritized, maintaining transparency and accountability to the affected communities.
- **Do No Harm:** The model must be designed to minimize False Negatives (missing a severely hungry household), as this error carries the highest humanitarian cost. Evaluation metrics will be weighted to prioritize Recall for the 'Severe hunger' class.

12. Conclusion

This project provides a robust and ethically grounded strategy for applying AI/ML to the critical challenge of humanitarian targeting in the Gaza Strip. By framing the problem as a **Multi-Class Classification** task and prioritizing model interpretability alongside accuracy, the resulting tool will enable aid organizations to move from broad distribution to precise, data-driven intervention. The successful implementation of this strategy promises to significantly enhance the efficiency of resource allocation, ensuring that aid reaches the households most desperately in need of support.

References

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